ASSIGNMENT-2

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Subject:- C.N

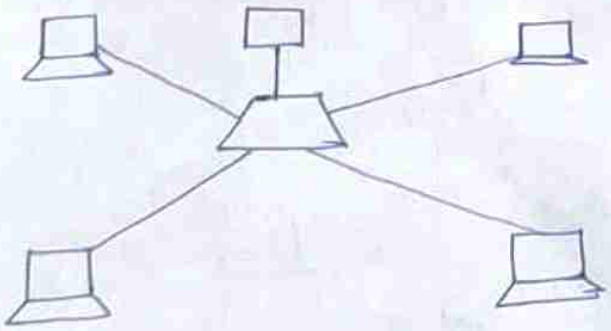
Q1) Explain the networking Topology.

Ans:- Network Topologies specifies the layout of a computer n/w. It shows how devices and cables are connected to each other.

★ Types of Topologies:-

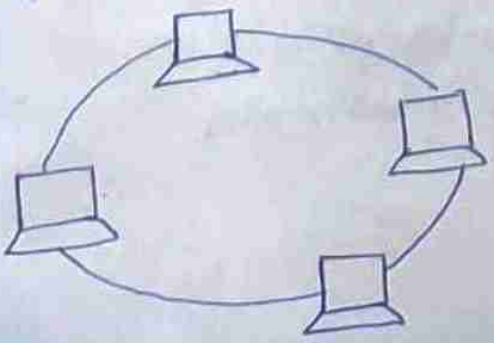
1) Star Topology:- It is a network topology in which all the nodes are connected to a single device known as a central device.

- It requires more cable compared to other topologies, therefore it is more robust as a failure in one cable.
- It is easy to install, manage and troubleshoot



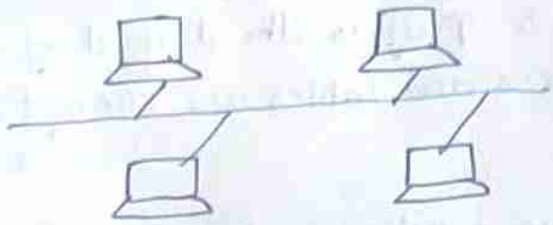
2) Ring Topology:- In a ring topology, the nodes are exactly con to two or more nodes and thus forming a single continu path for the transmission.

- Ring Topology does not need any central server to con the connectivity among the nodes
- If a single node is damaged, then whole network



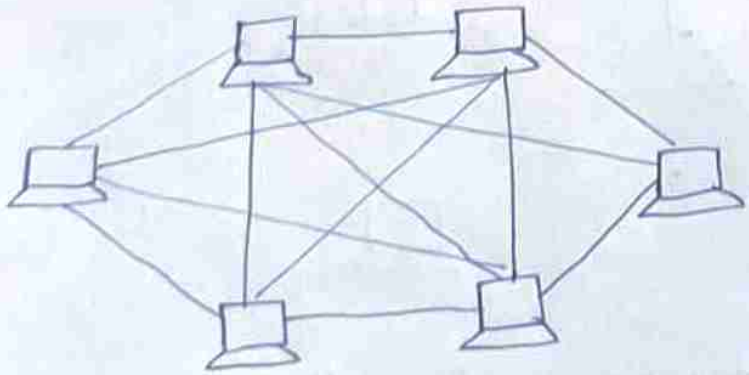
3) Bus Topology:- → All the nodes are connected to a single cable known as a central cable or Bus.

- It acts as a shared communication, i.e if any device wants to send data to other devices, then it will sends the data to all the attached devices.



4.) Mesh Topology:- In mesh topology all the nodes are individually connected to all other nodes.

- Does not need any central switch or Hub.
- It is categorised into 2 parts.
  1) Full-connected Mesh Topology
  2) Partially connected Mesh Topology.



5.) Tree Topology:- It is the combination of star and bus topology.

- Also known as expanded star topology.
- In tree topology all the star networks are connected to single bus.
- Ethernet protocol is used.
- If one segment is damaged, there is no eff on other segments.

6) Hybrid Topology:- Combination of different topology form a resulting topology.
- Provides flexibility as it can implemented in a different network environment.

Q20 Explain switching and its types:-

Ans:- Switching is a fundamental concept in networking and telecommunications that refers to process of directing data from one source to its destination across a network.

→ Switching determines the path that data packets take through the network to reach their destination.

→ There are 3 primary types of switching techniques:-

1.) ~~Network~~ Circuit Switching:-

→ Circuit switching involves establishing a dedicated communication path between the source and destination, before the actual data transmission begins.

→ The circuit remains reserved for the duration of the communication session.

eg Traditional Telephone network.

★ Characteristics:-

- Dedicated Path → A dedicated communication path is reserved for the centre session.
- Fixed Bandwidth → The resources (bandwidth) are reserved, ensuring a constant data rate.

★ Disadvantage:- Inefficient for data communication.

2.) Packet Switching:-

→ Packet Switching breaks down into smaller packets that sent independently through the network.

→ eg The internet & most modern data networks.

★ Characteristics:-

- No dedicated path → Resources are shared and data is sent in smaller packets over different routes.
- Efficient Bandwidth Usage:- N/W resources are used more efficiently

★ Disadvantages:-

→ Potential delays due to packet routing and reassembly.
→ Not ideal for real-time communication.

3.) Message Switching:-

→ In message switching, entire message are sent from source to destination without breaking them into packets.
→ The message is stored at intermediate nodes and forward to the next node in the path ~~one~~ one hop at a time.

eg early telegraph networks and ~~usage~~ message switching.

★ Characteristics:-

- Store & forward → Message are temporarily stored at each intermediate node and forwarded when the next node is available.

- Inefficient for Real time communication
- It can lead to significant delays.

- Disadvantages:-

→ Large message size can result in delays due to storage processing at intermediate nodes.